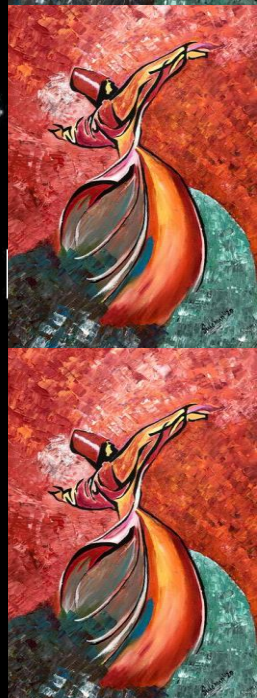




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



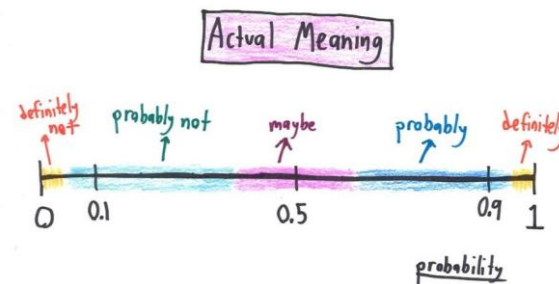


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 12 - Lecture No 22 – March 30, 2026



Announcements!

1. Office hours (post Ramadan) – Tuesday & Thursday 12:00 to 1:00 PM
2. Quiz No 06 – April 01, 2026 (Wednesday)
3. Planned quizzes
 1. Quiz No 07 – April 08, 2026 (Wednesday)
 2. Quiz No 08 – April 15, 2026 (Wednesday)
 3. Quiz No 09 – April 22, 2026 (Wednesday)
 4. Quiz No 10 – April 29, 2026 (Wednesday)

Agenda for today

1. Unit 5 – Continuous Random Variable

Example – Calculating Probability for a Standard Normal (Gaussian Random Variables)

$$X \sim N(0, 1)$$

1. find $P(X \geq 0)$?

$$P(X \geq 0) = 1/2$$

2. find $P(X \geq 10)$?

$$= \int_{10}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

No closed form available

3. How about $P(X \leq x)$?

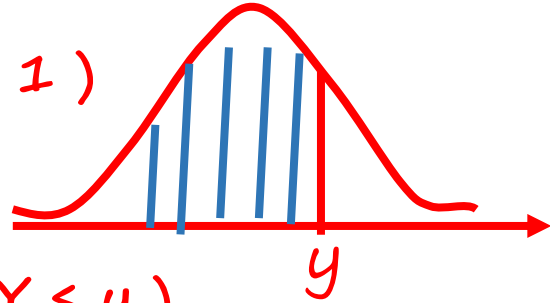
$$= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

No closed form CDF

Standard Normal Tables

- No closed form available for CDF
- But we have tables, for the standard normal

→ $Y = N(0, 1)$

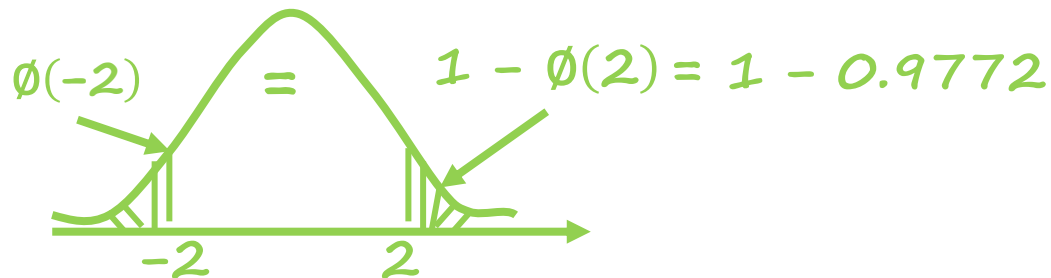


→ $\Phi(y) = F_Y(y) = P(Y \leq y)$

→ $\Phi(0) = P(Y \leq 0) = 0.5$

→ $\Phi(1.16) = 0.8770$

→ $\Phi(2.9) = 0.9981$

[illegible]

Example

Let's say $X \sim N(\mu_x, \sigma^2)$

find $P(X \geq \mu_x)$?

$$P(X \geq \mu_x) = 1/2$$

find $P(X \geq 10)$?

$$= \int_{10}^{\infty} \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} dx$$

No closed form available

How about $P(X \leq x)$?

$$= \int_{-\infty}^x \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} dx$$

No closed form CDF

What should we do?

$$X \sim N(\mu_x, \sigma^2)$$

$$P(X \leq x) = ?$$

$$\begin{aligned} P(X \leq x) &= P(X - \mu_x \leq x - \mu_x) \\ &= P\left(\frac{X - \mu_x}{\sigma_x} \leq \frac{x - \mu_x}{\sigma_x}\right) \\ &= P\left(Y \leq \frac{x - \mu_x}{\sigma_x}\right) \end{aligned}$$

$$\begin{aligned} \text{Where, } Y &= \frac{X - \mu_x}{\sigma_x} \\ &= \frac{1}{\sigma_x} X - \frac{\mu_x}{\sigma_x} \end{aligned}$$

$$\begin{aligned} E[Y] &= E\left[\frac{1}{\sigma_x} X - \frac{\mu_x}{\sigma_x}\right] \\ &= \frac{1}{\sigma_x} \mu_x - \frac{\mu_x}{\sigma_x} = 0 \end{aligned}$$

$$\text{Var}(Y) = \frac{1}{\sigma_x^2} \cdot \sigma_x^2 = 1$$

$$Y \sim N(0, 1)$$

Standardizing a random variable

- Let X have mean μ and variance $\sigma^2 > 0$

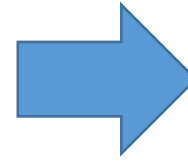
- Let $Y = \frac{X - \mu}{\sigma}$ $\Rightarrow E[Y] = 0$ $\Rightarrow \text{Var}[Y] = \frac{1}{\sigma^2} \text{Var}(X) = 1$

$$\Rightarrow X = \mu + \sigma Y$$

- Also, if X is normal, then: $Y \sim N(0, 1)$

Example (continued)

Let's say $X \sim N(\mu_x, \sigma^2)$



$$P(X \leq x_1) ?$$

$$P(Y \leq \frac{x_1 - \mu_x}{\sigma_x})$$

$$P(X \leq x) ?$$

$$= \int_{-\infty}^x \frac{1}{\sigma_x \sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} dx \Rightarrow P(Y \leq \frac{x - \mu_x}{\sigma_x})$$

$$\text{Where, } Y = \frac{X - \mu_x}{\sigma_x}$$

$$Y \sim N(0, 1)$$

Question No 03 [25 points]

X, Y are identical and independent Normal Standard RV's. Find the Probability (by showing calculations) for the following events (use the chart given on following page): -



(a) [5 points] $P(X < 2 \text{ or } X > 2)$

(b) [5 points] $P(X^2 < 4)$

(c) [5 points] $P(X < 2 \text{ and } Y > 3)$

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TABLE 5.1: AREA $\Phi(x)$ UNDER THE STANDARD NORMAL CURVE TO THE LEFT OF X

X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
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2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
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2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
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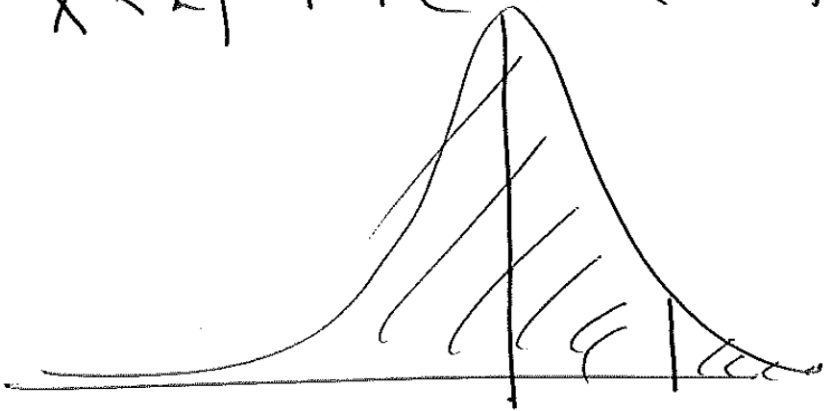
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(e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$\begin{aligned} \text{a)} \quad & P(X < 2 \text{ or } X > 2) \\ &= P(X < 2) + P(X > 2) \\ &= P(-\infty \leq X < 2) + P(2 \leq X \leq \infty) \\ &= 1 \end{aligned}$$


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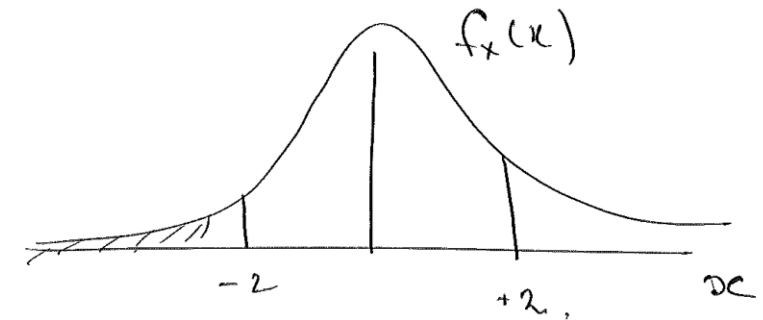
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(e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$(b) \quad P(X^2 < 4) \\ = P(-2 < X < 2)$$



$$\begin{aligned} &= P(X < 2) - P(X < -2) \\ &= P(X < 2) - (1 - P(X < 2)) \\ &= 2P(X < 2) - 1 \\ &= 2(0.9772) - 1 \\ &= 0.9544 \end{aligned}$$

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2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
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3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
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(e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$\begin{aligned} (c) \quad & P(X < 2 \text{ and } Y > 3) \\ &= P(X < 2) \cdot P(Y > 3) \\ &= P(X < 2) \cdot (1 - P(Y < 3)) \\ &= (0.9772) (1 - 0.9984) \\ &= 0.9772 \times 0.0016 \\ &= 0.00127036 \end{aligned}$$

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X	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Question No 03 [25 points]

X, Y are identical and independent Normal Standard RV's. Find the Probability (by showing calculations) for the following events (use the chart given on following page): -

(a) [5 points] $P(X < 2 \text{ or } X > 2)$

(b) [5 points] $P(X^2 < 4)$

(c) [5 points] $P(X < 2 \text{ and } Y > 3)$

➡ (d) [5 points] $P(XY < 0)$

(e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$\begin{aligned} \text{(d)} \quad P(XY < 0) &= P(X < 0) \cdot P(Y > 0) + P(X > 0) \cdot P(Y < 0) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

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➡ (e) [5 points] $P(X+5 < 0 \text{ and } Y+5 < 0)$

$$\begin{aligned} & P(X+5 < 0 \text{ and } Y+5 < 0) \\ &= P(X < -5) \cdot P(Y < -5) \\ &= 2P(X < -5) \\ &= 2(1 - P(X < 5)) \\ &= 2(1 - 0) \\ &\approx 0. \end{aligned}$$

Conditional PDF, given an event

$$P(A) > 0$$

$$p_X(x) = \mathbf{P}(X = x)$$

$$f_X(x)\delta \approx \mathbf{P}(x \leq X \leq x + \delta)$$

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A)$$

$$\Rightarrow f_{X|A}(x)\delta \approx \mathbf{P}(x \leq X \leq x + \delta \mid A)$$

$$\mathbf{P}(X \in B) = \sum_{X \in B} p_X(x)$$

$$\mathbf{P}(X \in B) = \int_B f_X(x) dx$$

$$\mathbf{P}(X \in B \mid A) = \sum_{X \in B} p_{X|A}(x)$$



$$\mathbf{P}(X \in B \mid A) = \int_B f_{X|A}(x) dx$$

Definition

$$\sum_x p_{X|A}(x) = 1$$

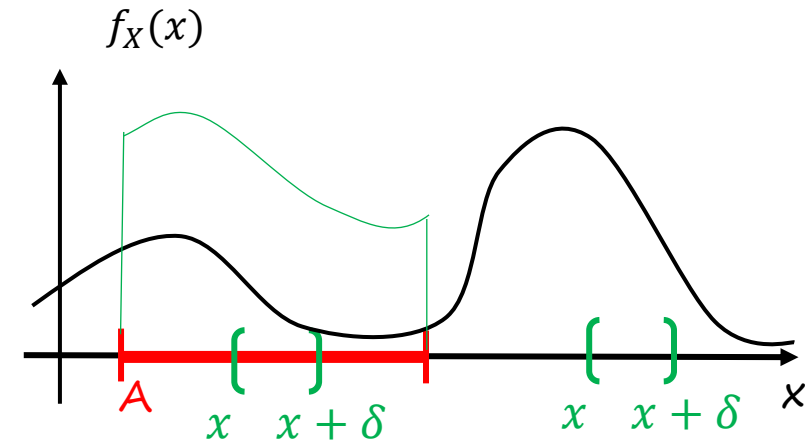
$$\int f_{X|A}(x) dx = 1$$

Conditional PDF of X , given that $X \in A$

$$\mathbf{P}(x \leq X \leq x + \delta \mid X \in A) \approx f_{X \mid X \in A}(x) \delta$$

$$\Rightarrow = \frac{P(x \leq X \leq x + \delta, X \in A)}{P(A)}$$

$$\Rightarrow = \frac{P(x \leq X \leq x + \delta)}{P(A)} \approx \frac{f_X(x) \delta}{P(A)}$$



$$f_{X \mid X \in A}(x) = \begin{cases} 0, & \text{if } x \notin A \\ \frac{f_X(x)}{P(A)}, & \text{if } x \in A \end{cases}$$

Conditional Expectation of X, given an event

$$E[X] = \sum_x x p_X(x)$$

$$E[X] = \int x f_X(x) dx$$

$$E[X | A] = \sum_x x p_{X|A}(x)$$



$$E[X | A] = \int x f_{X|A}(x) dx$$

Expected Value Rule

$$E[g(X)] = \sum_x g(x) p_X(x)$$

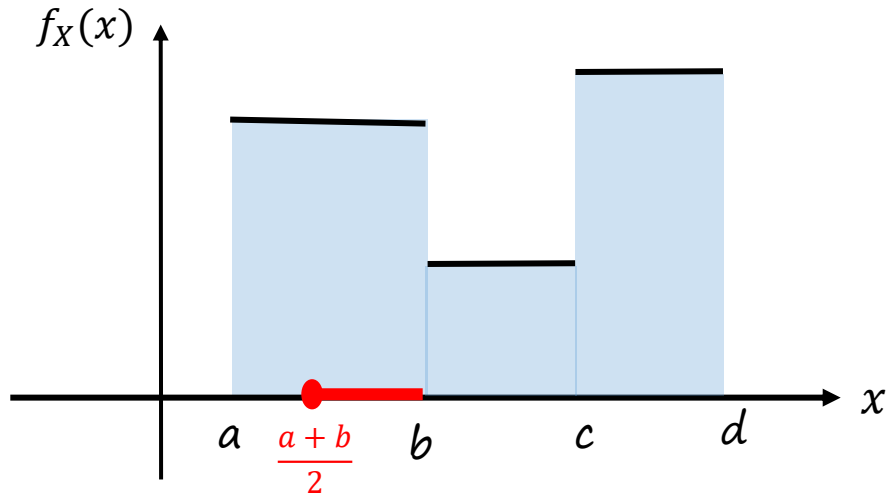
$$E[g(X)] = \int g(X) f_X(x) dx$$

$$E[g(X) | A] = \sum_x g(x) p_{X|A}(x)$$



$$E[g(X) | A] = \int g(X) f_{X|A}(x) dx$$

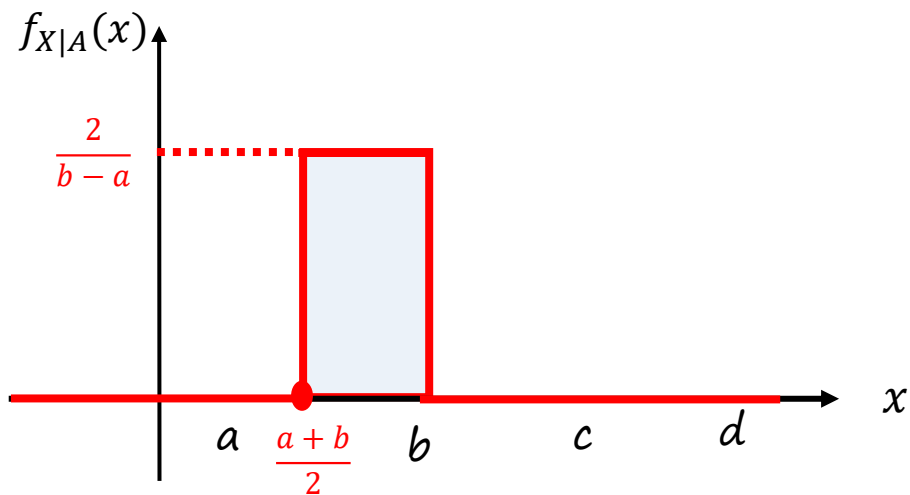
Example



$$A: \quad \frac{a+b}{2} \leq X \leq b$$

$$\begin{aligned} \Rightarrow E[X | A] &= \frac{1}{2} \frac{a+b}{2} + \frac{1}{2} b \\ &= \frac{1}{4} a + \frac{3}{4} b \end{aligned}$$

$$\Rightarrow E[X^2 | A] = \int_{\frac{a+b}{2}}^b \frac{2}{b-a} x^2 dx$$



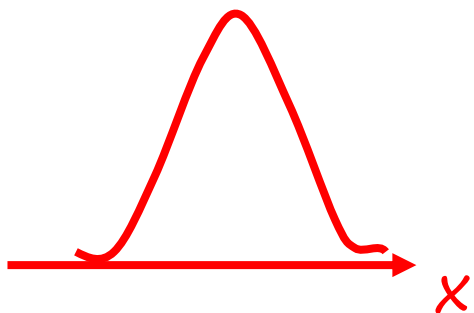
Exercise

Assume $X \sim N(0, 1)$

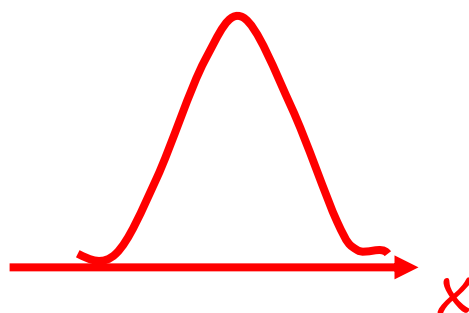
Draw $f_{X|A}(x)$

Where $A =$

a) $X \in \pm \sigma$



b) $X \in \pm \frac{\sigma}{2}$



c) $X \in \pm \frac{\sigma}{100}$

